

AHMED BARGOUGUI

AI & Data Engineering | Backend Developer | Computer Science — Final Year

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PROFILE

Final-year Computer Science student with two internships at Sagemcom — **Telecom AI Engineering (RAG, OCR, CrewAI, n8n automation, validation pipelines)** and Business Intelligence. Built production backend systems used by **32,000+ users** and AI-powered applications. Strong interest in **end-to-end AI pipelines, generative AI, and scalable backend architecture**. Seeking a PFE internship in AI or Backend Engineering.

PROFESSIONAL EXPERIENCE

Telecom Data & AI Engineering Intern — Sagemcom • Ben Arous, Tunisia 07/2026 – 08/2026

- Built **ICP/MTP validation** system (OCR extraction, PASS/FAIL compliance reports) + CrewAI Generative AI layer for automated interpretation
- Designed end-to-end **n8n + Gmail** automation: validation → Generative AI → production webhook → automated email reporting

Business Intelligence Engineer Intern — Sagemcom • Ben Arous, Tunisia 07/2025 – 08/2025

- Star Schema DWH + ETL pipelines (Pandas/SQLAlchemy) + **Power BI** dashboards for real-time KPI tracking; standardised 10,000+ records via Power Query

PROJECTS

AI Risk Analyst — Enterprise Risk Platform — github.com/Ahmedbrg/ai-risk-analyst 2026

- FastAPI + React platform with **CrewAI 5-agent** system, Qdrant RAG, deterministic ISO 31000-aligned scoring, n8n automation, and QC-gated PDF export
- Stack:** Python • FastAPI • CrewAI • Qdrant • RAG • React • n8n • Docker

ScanVD — AI Video Search Engine — github.com/Ahmedbrg/ScanVD • huggingface.co/spaces/AhmedBrg/ScanVD_demo2026

- Multi-model AI pipeline (**Whisper + YOLOv8 + CLIP**) extracting precise timestamps from natural language queries — deployed on HuggingFace
- Stack:** Python • Whisper • YOLOv8 • CLIP • HuggingFace • Docker

Forsa — Tunisian Mobile Super-App — github.com/Ahmedbrg/Forsa 2026

- Dark-themed **React Native super-app** combining community feed, marketplace, freelance jobs, and messaging — JWT-backed Express/MongoDB API
- Stack:** React Native • TypeScript • Expo • Node.js • Express • MongoDB • JWT • Zustand

Secure Auth System — Gaming Community — Freelance • Private 2026

- Secure REST API (Node.js, Express, MySQL) + C# desktop launcher for a **32,000-member** community — HWID binding, OAuth2, JWT, rate limiting
- Stack:** Node.js • Express • MySQL • C# • JWT • OAuth2

TECHNICAL SKILLS

AI / GenAI:	RAG 00B7 LLMs 00B7 CrewAI 00B7 Qdrant 00B7 Whisper 00B7 YOLOv8 00B7 CLIP 00B7 OCR 00B7 Embeddings
Automation:	n8n 00B7 Webhooks 00B7 Gmail API 00B7 Workflow Automation 00B7 JSON Pipelines
Backend:	Node.js 00B7 Express 00B7 FastAPI 00B7 REST APIs 00B7 JWT 00B7 OAuth2 00B7 MySQL 00B7 MongoDB 00B7 C#
Data / BI:	SQL Server 00B7 ETL 00B7 Pandas 00B7 SQLAlchemy 00B7 Power BI 00B7 Star Schema
Languages:	Python 00B7 JavaScript 00B7 TypeScript 00B7 Java 00B7 C++ 00B7 C# 00B7 SQL
Tools:	Git 00B7 Docker 00B7 Linux 00B7 Postman 00B7 VS Code 00B7 Jupyter

EDUCATION • LEADERSHIP • LANGUAGES

Bachelor's Degree in Computer Science — FSTSBZ, University of Kairouan 09/2024 – 06/2027

Final Year (PFE) • Algorithms • Data Structures • Databases • OOP • OS • Networks • AI • AgriLink: AI farming app (TGN Hackathon 2025, React Native)

- Founder & President, Club DevUp** — AI/ML & career workshops, 50+ students (11/2025–05/2026) • **Trainer & Jr. Secretary General, JCI Sidi Bouzid** (2023–Present)
- Ranked 3rd — ACM-style Competitive Programming Contest** • **Arabic** Native • **French** Professional • **English** Professional